

Determining Apartment Rental Prices

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Introduction

Determining the optimal renting price for apartments can be very challenging due to the number of potentially impactful variables. Using a dataset of 677 apartments with respect to several variables which will be elaborated upon in a later section, this report will deduce the most useful model in determining rent price, thereby allowing decisions to be made regarding the price of three newly constructed apartments.

It is necessary to use specialized statistical models to approach this problem because of the variables that are present. The “solution” and methods may be non-trivial due to potential interactions between said variables, further necessitating the use of these models. Computer programs, namely SAS and Stata, will facilitate the completion of linear regression analyses and estimations, greatly simplifying the complex processes. The analyses will be completed using the least squares method of residuals.

Dependent Variable:

The dependent variable for this support will be the rental price of the apartment, as it is the value to be predicted. The goal is to gain an understanding of which variables are most significant in explaining rental price.

y = rent in U.S. dollars

Independent Variables:

The following list includes the raw explanatory variables for the dependent variable. While this list is not exhaustive, they will likely provide an idea as to which factors significantly influence apartment rental price. Additional independent variables will be included in more complex models later in the report, along with their respective explanations and justifications.

x_1 = number of bedrooms

x_2 = number of bathrooms

x_3 = square footage of apartment

x_4 = indicator variable, 1 if apartment (0 otherwise)

x_5 = indicator variable, 1 if apartment complex (0 otherwise)

x_6 = indicator variable, 1 if house (0 otherwise)

x_7 = distance from school in feet

x_8 = indicator variable, 1 if apartment is in city A, 0 if apartment is in city L

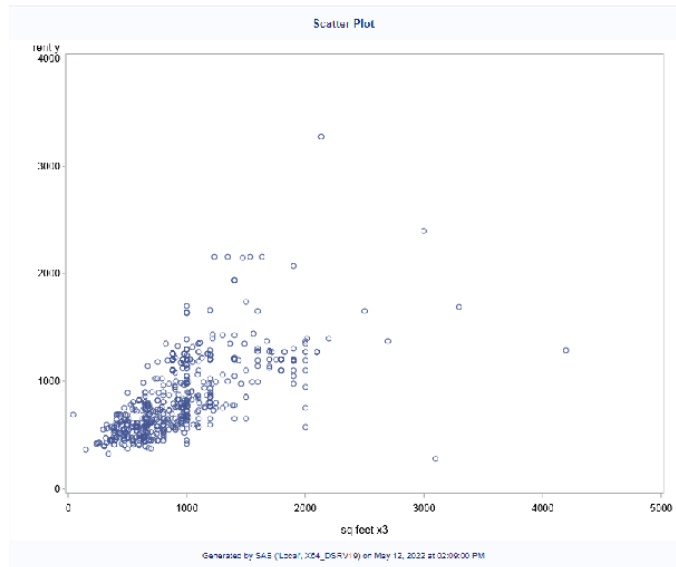
Variable Summaries:

The table to the right, created using Stata, includes the summary statistics for each of the independent variables. We immediately see that there is an extremely high standard deviation for square feet, meaning there is a great deal of variability in the size of rental buildings. It should also be noted that the means of the indicator variables represent the proportion of observations which are in that category. For example, about 78.3% of observations were in City A.

sum rent bedroom bathroom sqft apt apt_comp house distance city						
Variable	Obs	Mean	Std. dev.	Min	Max	
rent	677	880.9572	369.8657	275	3280	
bedroom	680	2.273529	1.362837	0	7	
bathroom	680	1.502941	.6392813	1	4	
sqft	613	936.0131	453.4315	42	4200	
apt	680	.4823529	.5000563	0	1	
apt_comp	680	.2191176	.413953	0	1	
house	680	.1191176	.3241653	0	1	
distance	680	1.804559	1.425972	.4	9.8	
city	680	.7838235	.411939	0	1	

Scatter Plots:

Before conducting a formal analysis on the data, it is useful to gain a preliminary understanding of the effects of major explanatory variables on rental price. The most trivial of these major variables would be square footage. One might expect rental prices to increase as square footage increases, as larger spaces are typically more expensive. The scatter plot to the right has square footage on the x-axis and rent price on the y-axis. The expected relationship is generally supported by the plot, in which the data appears to show an upward trend. However, there are several outliers, namely the points at approximately (3050, 200), and (4100, 1200). The existence of these outliers suggests that other variables must be included in the analysis. An attempt to generate least squares estimates for the above relationship would yield results that may be inadequate and inaccurate, reducing the usefulness of the model. Thus, the first linear regression model will include all of the independent variables, and be adjusted until it provides useful and satisfactory results. It should be noted that there are several assumptions which may be adhered to with linear regression models - explanations for these will be included.



The Newly Constructed Apartments:

Below are the three new apartments for which the report will make predictions. Each is listed with all of the aforementioned explanatory variables. This data will be used in the linear regressions to give estimates on what their ideal rental price would be. Note that additional explanatory variables will be included in more complicated models later in the report.

Unit	Bedrooms x_1	Bathrooms x_2	Sq Feet x_3	APT x_4	Complex x_5	House x_6	Distance x_7	City x_8
700	0	1	390	0	1	0	1.3	1
800	2	2	792	1	0	0	1.9	0
900	4	2	1000	1	0	0	0.6	1

Notes on Linear Regression Assumptions

Before conducting linear regression analyses, it is extremely important to consider the underlying assumptions for such tests. There are five major assumptions, all of which are explained in more detail in the following list:

- **Linearity** - This assumption is that the relationship between x and the mean of y is linear. In the case of this investigation, y is the rental price, and x is represented by the explanatory variables listed on the previous page. Though trivial, it should be noted that this assumption only applies to linear regression models.
- **Homoscedasticity** - For this assumption, it must hold that the variance of the residuals is the same for any value of x . That is, regardless of the values of the explanatory variables, the residuals will be constant. If this assumption is violated, it means that there is some pattern remaining in the data, implying that the model treats certain explanatory variable values differently than others, creating bias, thereby reducing the usefulness of the model.
- **Independence** - Each observation in the sample must be independent of each other. While this assumption can be difficult to verify, it can typically be concluded if each observation was selected at random. It might be likely for rental prices of certain apartments to have an effect on the prices for others, but this factor cannot necessarily be accounted for with the provided dataset. Therefore, it will be assumed that independence of observations is not violated in the sample. This will be the case for each model in the report. Additionally, each residual must be independent of all the others residuals.
- **Normality** - It must hold that the residuals produced by each model are approximately normally distributed
- **Multicollinearity** - There must be relatively little to no multicollinearity between independent variables. In other words, any two variables must not be too closely correlated. This will be addressed through the use of variance inflation factors.

Basic Model

General Equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + e$$

Rationale:

Before creating a complex model with interaction terms or quadratic terms, it is important to develop a relatively simple model. This way, we can determine which variables are least significant in predicting rent price and exclude them from future models, streamlining the process somewhat. The model will be created through SAS, utilizing the least squares method to create a line of best fit for the data.

Fitted regression line and SAS Output:

$$\begin{aligned} \hat{y} = & 222.523 + 149.765x_1 + 102.907x_2 \\ & + 0.114x_3 + 61.506x_4 + 17.419x_5 \\ & + 138.361x_6 - 45.336x_7 + 108.799x_8 \end{aligned}$$

Inferences, Interpretation, and Assumptions:

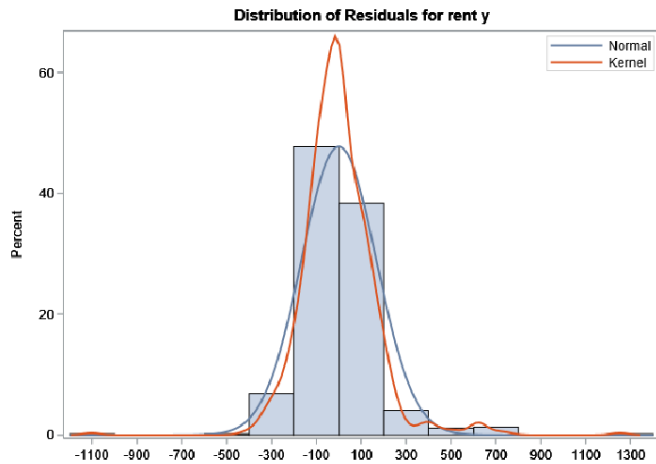
- The analysis of variance table gives a p-value less than 0.001, meaning that the model is useful in predicting rental prices at $\alpha = 0.05$
- According to their corresponding p-values, all variables are significant in predicting rent price except for the indicator variable for apartment complexes
- All variance inflation factors are less than 10, suggesting none of the variables are completely collinear with any of the others. In other words, there was no “duplicated” information in the data
- Distance appears to have a negative effect on rent price. That is, as distance from the school increases, apartment rent price decreases. This seems logical as proximity to important buildings is certainly a factor in rent pricing
- The parameter estimate for square feet is only 0.114, meaning if square footage increased by 1, then rent price will increase by \$0.114. This is a small value but it is logical as square footage would rarely increase by only 1 unit
- Looking strictly at the t-values from the table, it appears that the number of bedrooms is the most significant in predicting apartment rent price, with a value of 18.1. This value far exceeds the absolute value of any other parameter estimates.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	81140548	7842569	270.47	<.0001
Error	801	10862424	28257		
Corrected Total	809	78122973			

Root MSE	168.09802	R-Square	0.7826
Dependent Mean	883.93934	Adj R-Sq	0.7797
Coeff Var	19.45716		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	222.52325	29.95638	7.43	<.0001
bedroom x1	1	149.76485	8.27301	18.10	<.0001
bathroom x2	1	102.90710	14.75857	6.97	<.0001
sq feet x3	1	0.11413	0.02699	4.23	<.0001
APT x4	1	61.50641	19.44038	3.16	0.0016
APT comp x5	1	17.41889	21.95440	0.79	0.4278
house x6	1	138.36149	27.30646	5.07	<.0001
distance x7	1	-45.33563	6.34927	-7.14	<.0001
city_t x8	1	108.79909	20.10285	5.31	<.0001

- The adjusted $R^2 = 0.7797$, meaning 77.97% of the variability in rent price is explained by the variables in the model. This value is adjusted for the amount of variables present
- The Mean Squared Error of the model is 28257, which appears high, but cannot be placed into perspective until it is compared to the values generated by the other models
- The residual density plot is included to the right. One of the assumptions for linear regression is that the distribution of the residuals, or the errors made by the model, is approximately normal. For the simple model, the residuals appear to be mostly normally distributed, despite the very high peak toward the mean. Additional outputs from SAS support the normality assumption, but there may still be an underlying pattern within the residuals which might be affecting the results.



Hypothesis Test(s):

Since the only insignificant variable according to the initial regression was x_5 , we will conduct a hypothesis test regarding its associated slope parameter to affirm the result. The test will be conducted at $\alpha = 0.05$

$$H_0: \beta_5 = 0 \text{ vs } H_1: \beta_5 \neq 0$$

$$\text{Test Statistic: } \frac{17.41989}{21.9544} = 0.793$$

Rejection Region / Critical Values:

$$\pm t_{0.025, 670} = \pm 1.96$$

Conclusion: The test statistic does not fall into the rejection region defined above, so the null hypothesis is not rejected. Thus, it can be concluded that the value of β_5 is in fact 0 at the 95% confidence level. In terms of the problem, this means that apartment complexes are statistically insignificant in predicting rental price. That is, it is insignificant whether a rental unit is an apartment complex or not. Since the variable was shown to be statistically insignificant, it

can be removed from the model. The tables to the right are the SAS output of the reduced model, which do not include the variable for apartment complexes.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	61122759	8731823	309.21	<.0001
Error	602	17000214	28240		
Corrected Total	609	78122973			

Root MSE	168.04630	R-Square	0.7824
Dependent Mean	883.93934	Adj R-Sq	0.7799
Coeff Var	19.45117		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	226.59411	29.50469	7.68	<.0001
bedroom x1	1	149.88721	8.29903	18.13	<.0001
bathroom x2	1	103.48496	14.73805	7.02	<.0001
sq feet x3	1	0.11293	0.02694	4.19	<.0001
APT x4	1	52.35538	15.64496	3.35	0.0009
house x6	1	128.88750	24.53778	5.25	<.0001
distance x8	1	-43.83834	6.09090	-7.23	<.0001
city_t x9	1	110.08217	19.68636	5.60	<.0001

Point Estimation:

Using the fitted regression line obtained from the SAS output, one can generate predictions for the rental price of the three apartment units simply by substituting the values for each explanatory variable. Recall that the variable for apartment complex is not included based on results from SAS and the hypothesis test.

- Unit 700:

$$\hat{y} = 226.594 + 149.887(0) + 103.484(1) + 0.113(390) + 52.355(0) + 129.867(0) - 43.839(1.3) + 110.082(1) = \$427.2393$$

- Unit 800:

$$\hat{y} = 226.594 + 149.887(2) + 103.484(2) + 0.113(792) + 52.355(1) + 129.867(0) - 43.839(1.9) + 110.082(0) = \$791.8929$$

- Unit 900:

$$\hat{y} = 226.594 + 149.887(4) + 103.484(2) + 0.113(1000) + 52.355(1) + 129.867(0) - 43.839(0.6) + 110.082(1) = \$1282.2436$$

Interval Estimation:

Next, we will use the above point estimates to generate a 95% confidence interval for the true value of the rental price for each apartment. Note that if this process were to be repeated many times, 95% of the generated intervals would contain the true value. The following values were calculated using the output SAS, including the estimated variance of the entire model.

- Unit 700: $(427.2393 \pm 1.96(19.457)) = (389.104, 465.375)$

- Unit 800: $(791.8929 \pm 1.96(19.457)) = (753.757, 830.029)$

- Unit 900: $(1282.2436 \pm 1.96(19.457)) = (1246.108, 1322.379)$

The above intervals can be used to come up with a reasonable price for the new apartment units based on the data collected in the initial survey.

Conclusions:

This model, while leaving some to be desired in terms of R^2 and MSE , provides an elementary understanding of which variables are statistically significant in predicting rental price. While the R^2 of 0.7797 is relatively high when compared to most linear regressions, there is certainly a high probability that rental price is less “random” than the value suggests. It seems unlikely that any apartment unit or house would have its rental price influenced by randomness, rather it would be determined by specific factors and procedures. There must be a method of explaining more of the variance in rental price, meaning less of the data can be attributed to pure randomness. Thus, the next section will explore a more complicated model in the hopes of improving our understanding of apartment rental prices.

Interaction Term Model:

General Equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_1 x_3 \\ + \beta_{10} x_2 x_3 + \beta_{11} x_3 x_4 + \beta_{12} x_3 x_5 + \beta_{13} x_3 x_6 + \beta_{14} x_3 x_7 + \beta_{15} x_3 x_8 + \beta_{16} x_1 x_2 + e$$

Rationale:

The variable for square feet has many interaction terms because it likely has some relationship with how many bedrooms and bathrooms there are, the type of building it is, which city it is in, etc. There may also be a relationship between the number of bedrooms and bathrooms in the rental unit, so a corresponding variable will be included for this as well. The inclusion of these extra variables could provide us with a stronger understanding of the dataset, improving the predictions we are able to make.

Fitted Regression Line and SAS Output:

The image to the right is a screenshot of the SAS output for the interaction term created above. The image immediately below it shows the summary of backward elimination, in which the program ran several linear regressions. Each step represents a variable being added or removed from the regression, with the threshold p-value being 0.1. That is, if a variable had a p-value greater than 0.1, it would be dropped from the regression as it is relatively insignificant in predicting rental price. Note that only one variable was removed at a time, starting with the highest p-value and working its way downward. The removal of any variable affects the entire regression, and thus can have meaningful impacts on the p-values of other variables. Some of the variables in the regression, notably the interaction terms between square feet and house type, became more significant as the distance and bedroom variables were dropped, but still remained above the 0.1 significance threshold, and were thus dropped themselves.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	63333134	7037015	285.48	<.0001
Error	600	14789838	24650		
Corrected Total	609	78122973			

Root MSE	157.00233	R-Square	0.8107
Dependent Mean	863.93934	Adj R-Sq	0.8078
Coeff Var	18.17284		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation	95% Confidence Limits
Intercept	1	290.89966	46.62721	5.60	<.0001	0	189.32729 352.47204
bathroom x2	1	115.23906	48.83846	2.36	0.0189	24.08799	19.32395 211.15417
sq feet x3	1	0.26550	0.05881	4.53	<.0001	17.48928	0.16040 0.36081
APT x4	1	53.47091	14.45636	3.70	0.0002	1.27492	25.07969 81.86213
house x6	1	106.76005	23.81740	4.52	<.0001	1.50540	60.37724 153.14285
sqft*bedroom x11	1	0.05337	0.01545	3.45	0.0009	37.11595	0.02303 0.08371
sqft*bathroom x12	1	-0.15713	0.01960	-8.02	<.0001	19.39838	-0.19562 -0.11864
sqft*city x18	1	0.15082	0.01622	9.30	<.0001	1.78993	0.11897 0.18266
sqft*distance x19	1	-0.03451	0.00523	-6.59	<.0001	3.08690	-0.04479 -0.02423
bedroom*bathroom x21	1	57.16338	11.34554	5.04	<.0001	37.08228	34.88157 70.44518

Summary of Backward Elimination								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1		city_t x9	15	0.0000	0.8125	15.0406	0.04	0.8405
2		APT comp x5	14	0.0000	0.8124	13.1598	0.12	0.7298
3	multiunit x7		15	0.0000	0.8125	15.0406	0.12	0.7298
4		multiunit x7	14	0.0000	0.8124	13.1598	0.12	0.7298
5		distance x8	13	0.0001	0.8123	11.5395	0.38	0.5374
6		bedroom x1	12	0.0001	0.8122	9.9148	0.38	0.5396
7		sqft*apt x14	11	0.0004	0.8118	8.1081	1.20	0.2739
8		sqft*house x16	10	0.0003	0.8115	8.2142	1.11	0.2922
9		sqft*apt comp x15	9	0.0008	0.8107	8.6708	2.47	0.1187

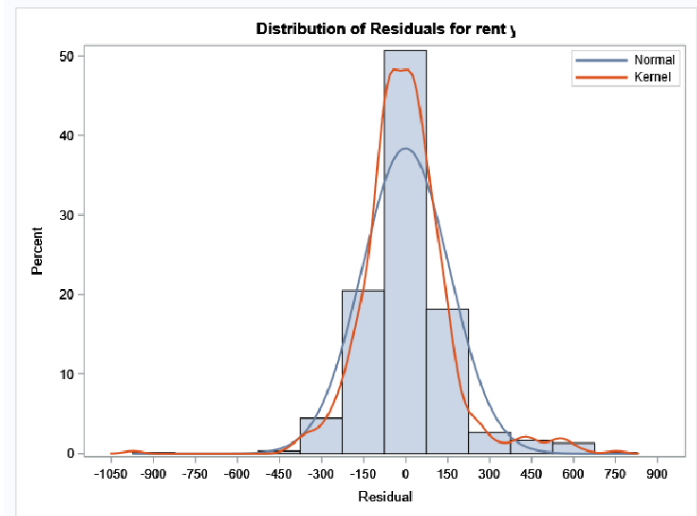
The fitted regression line generated from SAS is as follows:

$$\hat{y} = 280.89 + 115.24x_2 + 0.27x_3 + 53.47x_4 + 106.76x_6 + 0.05x_1x_3 - 0.17x_2x_3 - 0.03x_3x_7 + 0.16x_3x_8 + 57.16x_1x_2$$

It should be noted that several base variables, namely bedroom, city, and distance, are involved in interaction terms despite being dropped due to statistical insignificance.

Inferences, Interpretations, and Assumptions:

- To the right is the histogram of residuals for the regression. It can be seen that they are relatively close to a normal distribution, albeit with a higher peak once again. Despite the high peak, one can conclude that the residuals are sufficiently normally distributed, meaning the normality of errors assumption has not been violated
- The adjusted R^2 value for this model increased by nearly three percentage points compared to the original model. Specifically, the variables in this model explained 80.78% of the variability in rent price, meaning there is less “randomness” remaining with the use of this model compared to the basic model. Note that the basic model had an R^2 of 77.99%. Additionally, the mean square error decreased from 28240 to 24650 between the models. Both of these facts imply that the interaction model is better at explaining rental price.
- Several variables in this regression possessed high variance inflation factors, particularly the variables bathroom, square feet times bedroom, and bedroom times bathroom. This means that the variables have a relatively high level of collinearity with at least one of the other variables, which could suggest that some of the information was duplicated in the dataset. However, if these variables were discluded from the regression model, the adjusted R^2 would decrease dramatically, reducing the usefulness of the model.



Point Estimation:

The following estimates were created using the fitted regression line at the top of this page and the data for the new apartment units on page 3.

- Unit 700: \$482.32
- Unit 800: \$772.096
- Unit 900: \$1294.12

Interval Estimation:

The following confidence intervals were created at the 95% confidence level:

- Unit 700: $(482.32 \pm 1.96(18.173)) = (446.7, 517.939)$
- Unit 800: $(772.096 \pm 1.96(18.173)) = (736.477, 807.715)$
- Unit 900: $(1294.12 \pm 1.96(18.173)) = (1258.501, 1329.739)$

Once again, the above intervals can be used to determine appropriate prices for the new units based on data from the survey. That said, these confidence intervals are likely more useful because they were created using a model which explains more of the variance in rent price. That is, the above confidence intervals leave less to random chance than the original model.

Conclusions:

The interaction term model is more useful in predicting rental price as it has a higher adjusted R^2 , a lower mean squared error, and nine significant explanatory variables, suggesting it is more useful than the original model. That being said, there is still a great deal of variability within rental price which could be explained by the inclusion of second-order variables.

Second-order with Interaction Model:

General Equation:

$$y = \beta_0 + \beta_1 x_2 + \beta_2 x_3 + \beta_3 x_4 + \beta_4 x_6 + \beta_5 x_7 + \beta_6 x_8 + \beta_7 x_1 x_3 + \beta_8 x_2 x_3 + \beta_9 x_3 x_7 + \beta_{10} x_3 x_8 \\ + \beta_{11} x_1 x_2 + \beta_{12} x_3^2 + \beta_{13} x_7^2 + e$$

Rationale:

The inclusion of second-order terms is to potentially explain the remaining variance in rental price. There may be a curvilinear or parabolic relationship between some of the variables and rental price, such as square footage or distance, so corresponding variables will be included in this model. Recall that several base variables are not included in the above equation as they were previously shown to be insignificant predictors of rental price.

Fitted Regression Line and SAS Output:

- The model is useful according to the p-value in the top table of the image on the following page. That is, at least one of the variables included in the model are statistically significant predictors of rental price.
- The adjusted R^2 value is .8290, meaning 82.9% of the variance in rental price is explained by the independent variables. This is an increase from both of the first two models. The mean squared error is 21937, which is also an improvement from the previous models, immediately suggesting this model is more useful in predicting y
- All of the explanatory variables are significant at $\alpha = 0.1$, and nearly all are significant at $\alpha = 0.05$. In other words, they are all significant at the 90% confidence level.

- The parameter estimate for square footage squared is very low, but this is unsurprising because the values for the variable were extremely high compared to the values of the other variables.
- It can be argued that square footage, distance, and distance squared are the most significant variables in the regression because they have the largest absolute t-values. This suggests that square footage and distance are the most important factors in determining rental price.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	65048658	5003743	228.10	<.0001
Error	508	13074315	21937		
Corrected Total	609	78122973			

Root MSE	148.11087	R-Square	0.8328
Dependent Mean	603.93834	Adj R-Sq	0.8290
Coeff Var	17.14384		

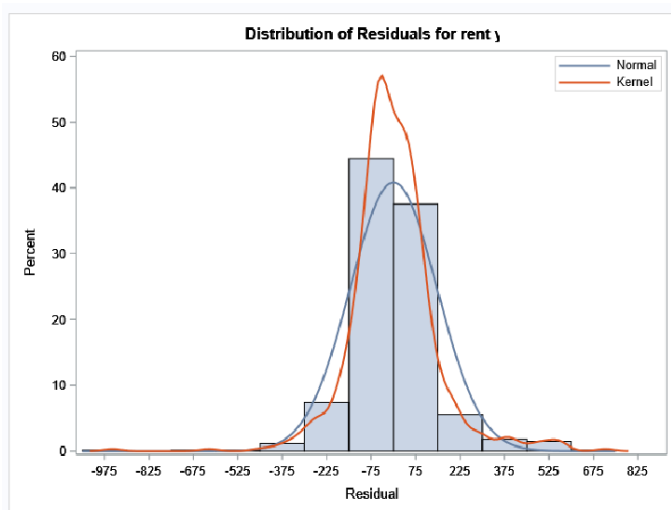
Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	408.75934	64.82198	6.33	<.0001	0
bathroom x2	1	99.84882	48.48094	2.08	0.0399	28.85091
sq feet x3	1	0.35958	0.07152	5.03	<.0001	29.26250
APT x4	1	34.74545	14.31770	2.43	0.0155	1.40524
house x6	1	45.62180	23.37138	1.95	0.0514	1.85881
distance x8	1	-122.38018	18.81012	-6.58	<.0001	20.51505
city_1 x9	1	-85.87417	40.91851	-2.10	0.0383	8.90825
sqft*bedroom x11	1	0.07864	0.01590	4.82	<.0001	44.16945
sqft*bathroom x12	1	-0.07189	0.03006	-2.39	0.0174	51.22680
sqft*distance x19	1	-0.04538	0.01158	-3.92	<.0001	18.98779
sqft*city x18	1	0.17831	0.03525	5.00	<.0001	9.50511
bedroom*bathroom x21	1	31.75583	11.78852	2.69	0.0073	44.57041
sqft*sqft x13	1	-0.00008186	0.00002471	-3.72	0.0002	29.67059
distance*distance x20	1	18.91706	2.26883	8.33	<.0001	12.58821

Fitted Regression Line:

$$\hat{y} = 408.759 + 99.849x_2 + 0.359x_3 + 34.745x_4 + 45.622x_6 - 122.39x_7 - 85.874x_8 + 0.077x_1x_3 - 0.072x_2x_3 - 0.045x_3x_7 + 31.756x_1x_2 - 0.00009x_3^2 + 18.917x_7^2$$

Inferences and Assumptions:

- Once again, the variance inflation factors for the variables are relatively high, meaning there is some level of multicollinearity between several variables. This was to be expected, as many of the included variables are based on others. While the VIFs are relatively high, none exceed 100, which is the general value used to determine variable exclusion. Further, the inclusion of said variables provides a more precise and useful explanation of rental price, so they should not be omitted.
- The residual histogram follows a similar pattern to the previous models in that it has a very high peak. However, it appears to remain approximately normal, meaning the assumption has not been violated.



Point Estimation:

Using the fitted regression line, we can again create estimates for the rental prices of the new apartment units.

- Unit 700: \$371.023
- Unit 800: \$774.054
- Unit 900: \$1150.752

Interval Estimation:

The following confidence intervals were created at the 95% confidence level:

- Unit 700: $(371.023 \pm 1.96(17.144)) = (337.421, 404.625)$
- Unit 800: $(774.054 \pm 1.96(17.144)) = (740.452, 807.656)$
- Unit 900: $(1150.752 \pm 1.96(17.144)) = (1117.149, 1184.354)$

Interestingly, the model predicted lower rent values for some of the new units and higher values for others. In the case of Unit 700, the resulting confidence interval is lower due to the parameter estimates with negative values. For Unit 800, the confidence interval results are extremely similar to those generated in the previous model, albeit narrower due to the lower model variance. Unit 900 was similar to Unit 700 in that its point estimate and interval estimates were both lower than previous models. This discrepancy in predictions is because of the specific parameter estimates, namely that for distance squared. It would be expected for this estimate to be negative, since the estimate for distance alone was negative. That is, one would expect the higher-order term to have the same sign as the corresponding lower-order term. Since Units 700 and 900 are closer to the school than Unit 800, their rent estimates are lower because the last term is not as large.

Hypothesis Test:

To determine whether this model is more useful than the previous model, an F test will be conducted. The quadratic model is denoted with subscript Q, and the simple interaction model with subscript I.

Hypotheses: $H_0: \beta_{12} = \beta_{13} = 0$, vs. $H_1: \text{At least one inequality}$

$$\text{Test Statistic: } \frac{SSR_Q - SSR_I}{2} * \frac{1}{MSE_F} = \frac{65048658 - 63333134}{2} * \frac{1}{21937} = 39.01$$

$$\text{Critical Value / Rejection Region: } F_{0.05, 2, 602} = 3.001$$

Conclusion: The test statistic is far greater than the critical value, so the null hypothesis is rejected. This means that the inclusion of the variables associated with β_{12} and β_{13} are statistically significant in predicting rental price. That is, the inclusion of square footage squared and distance squared is important in modeling the rent prices of apartments.

Conclusions:

This model had the highest adjusted R^2 value of any model thus far, along with the lowest mean squared error, suggesting it is the most useful of the three. Based on the results of the hypothesis test, it can be concluded that this model is more useful than the second model. That is, it was shown that the difference between the two models was statistically significant.

Overall Summary

Upon analysis of three separate regression models, it can be concluded that the third model is the most useful in predicting rental price because all of its variables are significant, it has the highest adjusted R^2 , and the lowest mean squared error. This is to say that the model can be used in determining rental prices for new apartment units. However, there is still a considerable amount of variability within the rental prices in the survey which could not be explained by the variables used in the model. With that in mind, it should be remembered that any of the three models explored in this report are entirely accurate or useful. That is, the models can be used as a tool in determining the prices, but are not perfect in doing so. It may be possible to explain more of the variance in rental price by including more variables, such as additional interaction terms or second-order terms, or perhaps combinations of the two. A potential downside of this would be the ever-increasing complexity of such a model, since more variables will make it more difficult to understand and thus more difficult to apply to real-world scenarios.

In terms of specific variables, it can be concluded that square footage, distance from the school, and city location of the apartment unit are very significant in determining renting price. While several of the other variables were shown to be significant as well, including number of bathrooms and building type, they were relatively insignificant compared to the aforementioned variables. That is, the size of the unit, its location, and proximity to the school are the most important factors in rental prices. It should also be noted that everything in this report is based upon one survey, and that results could vary greatly should the survey be repeated in a different area of the country or if it included additional explanatory variables. Such variables could include the company which owns the unit, or the company responsible for its construction, or potentially the floor of the unit.

Overall, it is very difficult to predict renting prices for buildings due to the great deal of factors which contribute to pricing. Despite this, the results of this report are adequately useful in understanding which factors are most influential in affecting rental prices, and can therefore be utilized in predicting future rental prices.